Heat-Kernel Boundary Sensitivity and a Two-Tier Infrared Basin for Lattice Yang–Mills

Research dossier

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Abstract

We implement the first theorem-level step of a representation-aware programme for the Yang–Mills mass gap. The heat-kernel plaquette action is selected as a primary regulator because gauge invariance, positive character coefficients, and convolution structure are explicit. We prove that a naive worst-case total-variation Dobrushin basin cannot be uniform in the small heat-time limit: on every nontrivial compact group, a heat kernel and any fixed nontrivial translate become asymptotically mutually singular. This is a constructive obstruction, not a no-go theorem for clustering. It shows that continuum basin entry must combine scale-sensitive good-field couplings with quantitative control of exceptional and topological blocks. We define such a two-tier basin, prove its abstract basin-to-clustering and stability theorems, give an executable $U(1)$ calibration, and isolate two explicit four-dimensional entry inequalities. Their verification for pure Yang–Mills remains open.

Contents

1 Target and result status **2**

2 Selection of the heat-kernel regulator **2**
2.1 Why this regulator is selected 3

3 The original basin and its hidden quantifier **3**

4 Compact-group heat-kernel obstruction **3**

5 Executable circle calibration **4**

6 Advanced regulator calibrations **5**
6.1 $SU(2)$ character calibration 5
6.2 Exact $U(1)$ convolution blocking 5
6.3 Defect-tail and regulator-comparison diagnostics 6

7 Two-tier good-field/defect basin **6**

8 Gauge-invariant good blocks and defect data **7**

9 Revised basin-entry theorem target **8**

10 Three implementation branches **8**
10.1 Influence branch 8
10.2 Polymer branch 9
10.3 Transfer branch 9

11 Relation to existing rigorous regimes **9**

12 Frontier audit **9**

13 Implemented diagnostics and immediate proof obligations **10**

1 Target and result status

The ultimate target is the construction of nontrivial four-dimensional pure Yang–Mills theory with a Hamiltonian gap above the vacuum for every compact simple gauge group [1]. The present paper addresses one earlier interface: a local, regulator-level criterion that could generate exponential infrared rigidity after renormalization.

The results are:

(i) a finite-lattice heat-kernel regulator and positivity certificate;

(ii) a compact-group obstruction theorem for worst-case total-variation influence;

(iii) a two-tier good-field/defect comparison theorem;

(iv) explicit replacement basin-entry inequalities; and

(v) tested $U(1)$ and $SU(2)$ heat-kernel calibrations, an exact $U(1)$ convolution-blocking experiment, and a curvature-matched Wilson comparison.

Items (ii)–(iii) are rigorous abstract results. The proof that the four-dimensional Yang–Mills RG trajectory satisfies the replacement inequalities is not supplied. Consequently no continuum construction or Clay-level mass gap is claimed.

2 Selection of the heat-kernel regulator

Let G be a compact connected Lie group with normalized Haar measure dg and bi-invariant Laplacian Δ_G. Let $p_t(g)$ be the heat kernel at time $t > 0$, normalized by

$$\partial_t p_t = \frac{1}{2}\Delta_G p_t, \qquad \int_G p_t(g)\, dg = 1.$$

It is central, satisfies $p_t(g^{-1}) = p_t(g)$, and has the character expansion

$$p_t(g) = \sum_{\lambda \in \widehat{G}} d_\lambda \mathrm{e}^{-tC_2(\lambda)/2} \chi_\lambda(g), \tag{1}$$

with positive coefficients. This action was introduced for lattice gauge theory by Menotti and Onofri [2].

On a finite oriented lattice $\Lambda = (V, E, P)$, assign $U_e \in G$ to each oriented edge with $U_{e^{-1}} = U_e^{-1}$. For an oriented plaquette p, let U_p be the ordered boundary holonomy. Define

$$d\mu_{t,\Lambda}(U) = Z_{t,\Lambda}^{-1} \prod_{p \in P} p_t(U_p) \prod_{e \in E^+} dU_e. \tag{2}$$

Proposition 2.1 (Finite gauge invariance PROVED)**.** *The measure* (2) *is invariant under* $U_{xy} \mapsto g_x U_{xy} g_y^{-1}$ *for arbitrary* $(g_x)_{x \in V} \in G^V$.

Proof. Plaquette holonomy transforms by conjugation at its base point. The heat kernel is central, and Haar measure is left- and right-invariant. □

Proposition 2.2 (Finite reflection-positive factor PROVED)**.** *Consider a lattice reflection that splits every plaquette crossing the reflection plane into a negative and a positive half-holonomy. Each crossing heat-kernel factor has a positive sum-of-squares decomposition. Consequently, with reflection-compatible boundary conditions, the standard finite-lattice reflection-positivity form is nonnegative.*

Proof. Write a crossing holonomy as $a^{-1}b$, where reflection exchanges the half variables. By (1),

$$p_t(a^{-1}b) = \sum_{\lambda,i,j} d_\lambda \mathrm{e}^{-tC_2(\lambda)/2} \overline{\lambda(a)_{ij}} \lambda(b)_{ij}.$$

The coefficient of every matrix element is nonnegative. Products of crossing factors retain a Gram decomposition; plaquettes away from the plane occur in reflected pairs. Integrating a positive-time cylinder function against its reflection therefore gives a sum of squared norms. □

This proposition is a finite-regulator certificate. RG blocking must either preserve the factorization or carry an independent recovery certificate.

2.1 Why this regulator is selected

Feature	Advantage	Remaining obligation
central heat kernel	exact gauge invariance	global orbit and topology control
positive characters	transparent finite positivity	preservation or recovery under RG
semigroup convolution	natural block comparison in two dimensions and toy sectors	no exact four-dimensional RG closure is assumed
small-time local Gaussianity	connects to Lie-algebra weak fields	worst-case boundary sensitivity degenerates
strictly positive density	finite conditional kernels are well defined	uniform physical-scale inequalities remain open

The Wilson regulator is retained as an adversarial comparison, not as a foundation. A universality theorem between the two remains part of the final continuum programme.

3 The original basin and its hidden quantifier

For a Gibbs specification (γ_i) on blocks $i \in I$, the classical worst-case total-variation coefficient is schematically

$$c_{ij} = \sup_{x=z \text{ off } j} \mathrm{TV}(\gamma_i(\cdot \mid x), \gamma_i(\cdot \mid z)) \,. \tag{3}$$

A uniform bound $\sup_i \sum_j c_{ij} < 1$ implies uniqueness and comparison estimates in its domain of applicability. Generalized block comparison theorems can extend that domain [3].

The critical word in (3) is "sup". In a weak-field limit the conditional law can become extremely narrow. Boundary data that move its center by a fixed group element then produce nearly disjoint conditionals even when typical neighboring configurations are much more coherent.

4 Compact-group heat-kernel obstruction

We use the convention

$$\mathrm{TV}(\mu,\nu) = \sup_A |\mu(A) - \nu(A)| = \frac{1}{2}\|\mu - \nu\|_{L^1}$$

when densities exist.

Lemma 4.1 (Disjoint concentration PROVED)**.** *Let X be a Hausdorff space and let $\mu_t \Rightarrow \delta_x$ and $\nu_t \Rightarrow \delta_y$ weakly as $t \downarrow 0$, with $x \neq y$. Then $\mathrm{TV}(\mu_t, \nu_t) \to 1$.*

Proof. Choose disjoint open neighborhoods $U \ni x$ and $V \ni y$ whose closures remain disjoint. By the Portmanteau theorem, after shrinking if necessary, $\mu_t(U) \to 1$ and $\nu_t(U) \to 0$. Thus $\mathrm{TV}(\mu_t, \nu_t) \geq |\mu_t(U) - \nu_t(U)| \to 1$, while total variation is at most one. □

Theorem 4.2 (Translated heat-kernel obstruction PROVED)**.** *Let G be a nontrivial compact Lie group, let $\mu_t(dg) = p_t(g)dg$, and fix $h \neq e$. If $L_h\mu_t$ denotes its left translate, then*

$$\lim_{t \downarrow 0} \mathrm{TV}(\mu_t, L_h\mu_t) = 1.$$

Proof. The heat kernels form an approximate identity, hence $\mu_t \Rightarrow \delta_e$. Translation is continuous, so $L_h\mu_t \Rightarrow \delta_h$. Apply Lemma 4.1. □

Corollary 4.3 (Failure of uniform worst-case contraction FALSIFIED)**.** *Consider the one-plaquette conditional family*

$$\gamma_t^\eta(dx) = p_t(x\eta^{-1})\, dx, \qquad \eta \in G.$$

Then

$$c(t) := \sup_{\eta, \eta'} \mathrm{TV}(\gamma_t^\eta, \gamma_t^{\eta'}) \longrightarrow 1 \quad (t \downarrow 0).$$

In particular, there is no $q < 1$ such that $c(t) \leq q$ for every sufficiently small t.

Proof. Choose $\eta' = h\eta$ with fixed $h \neq e$. The two conditional laws are translates covered by Theorem 4.2. □

Remark 4.4. The theorem does not prove absence of a mass gap, uniqueness, or clustering. It falsifies a proposed proof resource: a cutoff-uniform worst-case single-plaquette total-variation coefficient. Larger blocks, quotient metrics, typical-set restrictions, and defect expansions remain available.

5 Executable circle calibration

For $G = U(1)$ with angle θ, the heat kernel is

$$p_t(\theta) = \frac{1}{2\pi}\left(1 + 2\sum_{n \geq 1} \mathrm{e}^{-n^2 t/2} \cos(n\theta)\right). \tag{4}$$

The supplied program truncates this positive character series at a certified coefficient tolerance and applies periodic trapezoidal quadrature.

t	$\mathrm{TV}(p_t, p_t(\cdot - \pi))$	$\mathrm{TV}(p_t, p_t(\cdot - 0.5\sqrt{t}))$	Gaussian limit
1.00	0.767544967	0.194274584	0.197412651
0.50	0.947357851	0.197387803	0.197412651
0.20	0.999111867	0.197412653	0.197412651
0.10	0.999998642	0.197412651	0.197412651
0.05	1.000000000	0.197412642	0.197412651
0.02	1.000000000	0.197412640	0.197412651

The final column is

$$\mathrm{TV}(N(0,1), N(0.5,1)) = \operatorname{erf}\left(\frac{0.5}{2\sqrt{2}}\right).$$

The fixed-shift computation illustrates Theorem 4.2. The $\sqrt{t}$ shift shows that merely rescaling the metric does not automatically produce a small coefficient: one obtains a nontrivial tangent-Gaussian influence. Actual contraction must come from block dynamics or a smaller response of the conditional center to boundary perturbations.

Seven unit tests check positivity at the calibration grid, zero-shift identity, translation symmetry, fixed-shift degeneration, the tangent-Gaussian limit, output schema, and fail-closed parameter validation.

6 Advanced regulator calibrations

6.1 $SU(2)$ character calibration

For $SU(2)$, write the conjugacy angle as $\theta \in [0, \pi]$. With the Casimir convention used by the executable artifact,

$$p_t^{SU(2)}(\theta) = \sum_{n \geq 0} (n+1) \mathrm{e}^{-tn(n+2)/2} \frac{\sin((n+1)\theta)}{\sin\theta}, \qquad dg_{\mathrm{class}} = \frac{2}{\pi} \sin^2\theta \, d\theta. \tag{5}$$

Translation by the nontrivial central element $-I$ sends $\theta \mapsto \pi - \theta$. Midpoint Haar quadrature gives

t	$\mathrm{TV}(p_t^{SU(2)}(\theta), p_t^{SU(2)}(\pi-\theta))$
0.50	0.770181734
0.20	0.991849661
0.10	0.999976815
0.05	1.000000000

This completes the simple-group numerical calibration of Theorem 4.2; the theorem itself does not depend on the truncation.

6.2 Exact $U(1)$ convolution blocking

The circle heat kernel obeys $p_t^{*m} = p_{mt}$. Therefore a one-dimensional chain or exactly convolutional plaquette block provides a controlled diagnostic of how fixed-shift sensitivity changes under coarse graining:

Steps m	coarse time mt for $t = 0.02$	antipodal TV
1	0.02	1.000000000
2	0.04	1.000000000
5	0.10	0.999998642
10	0.20	0.999111867
20	0.40	0.973991076
50	1.00	0.767544967

Blocking reduces the fixed-shift coefficient only after the accumulated heat time becomes appreciable. Four-dimensional gauge blocking is not an exact plaquette convolution, so this is a mechanism calibration rather than an RG theorem.

6.3 Defect-tail and regulator-comparison diagnostics

Define a circle good field by $|\theta| \le 2\sqrt{t}$. Let R_{tail} be its complement probability and let $K_{0.25}$ be the TV response to a $0.25\sqrt{t}$ boundary shift. The table also compares the heat kernel with the curvature-matched Wilson density $Z^{-1}\exp(t^{-1}\cos\theta)$.

t	HK/Wilson TV	R_{tail}	$K_{0.25}$	$8(K_{0.25}+R_{\mathrm{tail}})$
0.50	0.063594784	0.045482439	0.099465621	1.159584482
0.20	0.019917856	0.045528368	0.099476451	1.160038549
0.10	0.009231571	0.045524826	0.099476455	1.160010248
0.05	0.004476830	0.045528348	0.099476460	1.160038463

The decreasing second column is a one-plaquette comparison, not a universality proof. The final column is deliberately labeled an *optimistic proxy*: it treats a raw defect probability as though it were the required Lipschitz defect influence and uses degree eight as a four-dimensional nearest-neighbor stress test. Its value above one shows that this toy choice of good radius and local shift does not enter even the replacement basin. A successful argument must gain contraction through a genuine block quotient coupling, a sharper defect-disagreement estimate, or both.

Together the basic and advanced suites contain fourteen passing tests.

7 Two-tier good-field/defect basin

The obstruction suggests splitting two effects that the worst-case supremum mixes together: smooth response inside a regular orbit chart and propagation through exceptional or topological blocks.

Let I be a finite or countable block graph with metric d. Let each block state space (S_i, ρ_i) have diameter at most one; for gauge systems it is a quotient or pseudometric on physical block data. Let $\gamma_i(\cdot \mid x)$ be local conditional kernels. Choose gauge-invariant good events G_i.

Definition 7.1 (Two-tier local coupling certificate)**.** *For boundary configurations x, z differing only at block j, a coupling $Q_{ij}^{x;z}$ of the two i-conditionals has a certificate (K_{ij}, R_{ij}) if*

$$\mathbb{E}_Q[\rho_i(X_i, Z_i)\,; X_i, Z_i \in \mathsf{G}_i] \le K_{ij}\rho_j(x_j, z_j), \tag{6}$$

$$\mathbb{E}_Q[\rho_i(X_i, Z_i)\,; \{X_i, Z_i\} \not\subset \mathsf{G}_i] \le R_{ij}\rho_j(x_j, z_j). \tag{7}$$

Here K measures regular transport and R dominates defect activation, topological mismatch, and chart failure. Both are local quantities; neither is defined by long-distance correlation decay.

The estimates imply a Wasserstein influence matrix $M \le K + R$. In a concrete construction, (7) can follow from a disagreement-polymer estimate rather than pointwise convexity.

Definition 7.2 (Two-tier basin)**.** *For $\kappa > 0$ and $q < 1$, a specification lies in $\mathfrak{B}_{2\mathrm{t}}(\kappa, q)$ when it has a two-tier certificate and*

$$\|K + R\|_\kappa := \sup_i \sum_j \mathrm{e}^{\kappa d(i,j)}(K_{ij} + R_{ij}) \le q < 1, \tag{8}$$

uniformly in total volume.

Lemma 7.3 (Certificate combination PROVED)**.** *The full coupling satisfies*

$$W_{1,\rho_i}(\gamma_i(\cdot \mid x), \gamma_i(\cdot \mid z)) \le (K_{ij} + R_{ij})\rho_j(x_j, z_j).$$

Proof. Add (6) and (7); together their events partition the coupling space. The Wasserstein distance is bounded by the cost of this coupling. □

Theorem 7.4 (Two-tier basin-to-clustering CONDITIONAL)**.** *Assume the local specification admits the Wasserstein comparison theorem for the seminorms*

$$\delta_i F = \sup_{x=z \text{ off } i} \frac{|F(x) - F(z)|}{\rho_i(x_i, z_i)}.$$

If it lies in $\mathfrak{B}_{2t}(\kappa, q)$*, then for local* F, G*,*

$$|\operatorname{Cov}(F, G)| \le \frac{C_0}{1-q} \mathrm{e}^{-\kappa d(S_F, S_G)} \left(\sum_i \delta_i F\right) \left(\sum_j \delta_j G\right), \tag{9}$$

where C_0 *is the normalization constant in the chosen comparison theorem.*

Proof. Let $M = K + R$. The weighted row norm is submultiplicative, so $D = (I - M)^{-1} = \sum_{n \ge 0} M^n$ and $\|D\|_\kappa \le (1-q)^{-1}$. Hence

$$D_{ij} \le (1-q)^{-1} \mathrm{e}^{-\kappa d(i,j)}.$$

The Wasserstein comparison theorem bounds covariance by $C_0 \sum_{ij} \delta_i F D_{ij} \delta_j G$. Nonzero seminorms lie in S_F and S_G, which yields (9). □

The comparison-theorem hypothesis is explicit because Gauss constraints can prevent a naive product-site theorem from applying directly. One must either formulate the blocks on physical quotient data or prove the constrained version required by the chosen regulator.

Proposition 7.5 (Openness and perturbative stability PROVED)**.** *Suppose* (K, R) *satisfies* (8) *with margin* $1 - q$ *and another effective theory has certificates* (K', R') *with*

$$\|(K' - K) + (R' - R)\|_\kappa \le \varepsilon < 1 - q.$$

Then it lies in $\mathfrak{B}_{2t}(\kappa, q + \varepsilon)$*.*

Proof. Apply the triangle inequality for the weighted row norm. □

Corollary 7.6 (Physical decay rate CONDITIONAL)**.** *If the basin is reached at physical block length* ℓ_* *and observable pullback errors are summable, the corresponding original observables have correlation rate at least* κ/ℓ_**. If the controlled separation is Euclidean time and OS reconstruction plus observable density hold, the positive spectral argument yields the same lower bound for the Hamiltonian gap.*

8 Gauge-invariant good blocks and defect data

For a block B, a regulator-level good event can be defined without a global gauge slice by

$$\mathsf{G}_B(\rho) = \left\{ U : \sum_{p \subset B} d_G(U_p, e)^2 \le \rho^2 |B| \right\}, \tag{10}$$

supplemented by physical bounds on boundary holonomy and an atlas label when a single orbit chart is insufficient. Plaquette conjugation invariance makes (10) gauge invariant.

The defect tier records at least:

- high-curvature blocks violating (10);
- failure of the chosen orbit chart or large transition functions between charts;
- nontrivial center flux, global holonomy, or bundle-sector mismatch;
- instanton-like concentration relevant at the selected scale; and
- RG remainder events whose reconstruction error exceeds its budget.

A volume-small probability is insufficient. What is required is a connected polymer estimate whose entropy is dominated uniformly in volume and across the number of RG steps used to reach ℓ_*.

9 Revised basin-entry theorem target

Let a be the microscopic spacing, $b > 1$ the block factor, and $\ell_k = b^k a$. Let $K^{(k)}$ and $R^{(k)}$ be the regular and defect matrices for the effective heat-kernel representation after k controlled steps.

Frontier obligation 9.1 (Good-field contraction $\mathrm{BE_G}$ OPEN)**.** *Prove that for some finite physical ℓ_* and $\kappa > 0$,*

$$\sup_B \sum_C \mathrm{e}^{\kappa d(B,C)} K_{BC}^{(k_*)} leq_G, \tag{11}$$

uniformly in cutoff and volume. The matrix must be obtained from couplings on gauge-orbit block data and must include all generated quasilocal interactions.

Frontier obligation 9.2 (Defect subcriticality $\mathrm{BE_D}$ OPEN)**.** *Construct a defect-disagreement matrix or connected-polymer domination with*

$$\sup_B \sum_C \mathrm{e}^{\kappa d(B,C)} R_{BC}^{(k_*)} leq_D, \qquad q_G + q_D < 1, \tag{12}$$

uniformly in cutoff and volume, including global orbit and topological data.

Theorem 9.3 (Revised conditional entry theorem CONDITIONAL)**.** *If $\mathrm{BE_G}$ and $\mathrm{BE_D}$ hold, the effective theory lies in $\mathfrak{B}_{2\mathrm{t}}(\kappa, q_G + q_D)$ and therefore satisfies Theorem 7.4.*

Proof. Combine (11) and (12) by the triangle inequality and apply the definition of the two-tier basin. □

This theorem is logically complete; its two hypotheses are the unresolved four-dimensional analytic work.

10 Three implementation branches

10.1 Influence branch

Construct a block conditional transport map on the quotient by internal gauge transformations. On G_B, derive a gauge-orthogonal Hessian or geodesic convexity estimate and bound the derivative of the conditional center with respect to physical boundary data. The target is (11), not mere convexity of a sampling dynamics.

10.2 Polymer branch

Decompose the full measure into good blocks and connected defects. Establish activities $z_k(X)$ satisfying, for a scale-independent $\alpha > 0$,

$$\sup_B \sum_{X \ni B} |z_k(X)| e^{\alpha |X|} \le \eta < 1. \tag{13}$$

Then construct $R^{(k)}$ from polymers connecting B to C. The key burden is to obtain (13) along the weak-coupling trajectory and through the finite physical entry scale, not only in the strong-coupling phase.

10.3 Transfer branch

Use Proposition 2.2 to construct the physical finite-lattice transfer operator. Seek a block-cone or projective contraction after quotienting gauge null directions. Any contraction constant must be uniform in volume and convert to physical units before it can enter a spectral conclusion.

These branches share the same defect and reconstruction ledger. Running them independently provides adversarial cross-checks: a local coupling estimate that loses positivity cannot be repaired by calling it a transfer estimate, and a polymer estimate restricted to the trivial orbit chart does not control the full measure.

11 Relation to existing rigorous regimes

Balaban's four-dimensional RG analysis supplies detailed small-field coupling renormalization but explicitly restricts that stage to a small-field approximation [4]. Strong-coupling lattice results can obtain functional inequalities and exponential decay in explicit regimes [5]. These are important calibrations on opposite sides of the programme:

- the former informs the good-field multiscale construction;
- the latter shows that local coupling and functional-inequality methods can yield regulated infinite-volume decay; and
- neither result proves (11)–(12) along the four-dimensional continuum trajectory.

The replacement basin is designed precisely to expose the bridge between those regimes: regular weak-field transport plus full-measure defect subcriticality at a finite physical scale.

12 Frontier audit

ID	Statement	Status	Claim level
HK1	finite heat-kernel gauge invariance and reflection-positive factor	PROVED	C2 regulated
HK2	fixed-translate heat-kernel TV tends to one	PROVED	C2
HK3	naive worst-case TV basin is cutoff-uniform	FALSIFIED	rejected
TB1	two-tier local certificate implies a Wasserstein influence bound	PROVED	C2
TB2	weighted two-tier basin implies clustering	CONDITIONAL	C2–C3
BE-G	four-dimensional good-field block contraction	OPEN	frontier

ID	Statement	Status	Claim level
BE-D	four-dimensional defect/topology subcriticality	OPEN	frontier
CONT	nontrivial regulator-independent OS-positive continuum theory	OPEN	frontier

The strongest advance is not a mass-gap proof. It is a rigorous elimination of an over-strong basin and a sharper decomposition of the first missing nonperturbative theorem. The new target is both more plausible and more demanding: it must show that regular response and exceptional propagation fit inside one contraction budget at a finite physical scale.

13 Implemented diagnostics and immediate proof obligations

N1. PROVED The $SU(2)$ character-series translation calibration is implemented and normalized against Haar measure.

N2. PROVED Exact $U(1)$ convolution blocking is implemented. The next step is a genuinely interacting two-plaquette or 2^4 block conditional kernel on quotient boundary data.

N3. OPEN Test whether quotient-Wasserstein response decreases with interacting block size after removing pure gauge modes.

N4. PROVED A one-plaquette good-field tail is implemented. The open theorem is a connected, boundary-uniform large-deviation bound for (10) with topology recorded.

N5. PROVED The first finite-scale optimistic $K + R$ proxy is measured and fails the degree-eight threshold. Replace the proxy by a certified block coupling and defect-disagreement matrix.

N6. PROVED A curvature-matched $U(1)$ one-plaquette heat-kernel/Wilson comparison is implemented. Extend it to common block observables with volume- and cutoff-uniform errors.

References

[1] A. Jaffe and E. Witten, *Quantum Yang–Mills Theory*, Clay Mathematics Institute problem description, `https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf`.

[2] P. Menotti and E. Onofri, "The Action of $SU(N)$ Lattice Gauge Theory in Terms of the Heat Kernel on the Group Manifold," *Nuclear Physics B* **190** (1981), 288–300.

[3] P. Rebeschini and R. van Handel, "Comparison Theorems for Gibbs Measures," *Journal of Statistical Physics* **157** (2014), 234–281, `https://web.math.princeton.edu/~rvan/dobrushin130819.pdf`.

[4] T. Bałaban, "Renormalization Group Approach to Lattice Gauge Field Theories I: Generation of Effective Actions in a Small Field Approximation and a Coupling Constant Renormalization in Four Dimensions," *Communications in Mathematical Physics* **109** (1987), 249–301, `https://projecteuclid.org/euclid.cmp/1104116842`.

[5] H. Shen, R. Zhu, and X. Zhu, "A Stochastic Analysis Approach to Lattice Yang–Mills at Strong Coupling," *Communications in Mathematical Physics* **400** (2023), 805–851, https://arxiv.org/abs/2204.12737.